
Energy-Adaptive FlashAttention: Joule-Efficient Attention via Runtime Power and Precision Control

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Despite record FLOP utilization on NVIDIA Hopper, state-of-the-art FlashAt-
2 tention kernels remain energy-inefficient relative to GEMM or cuDNN SDPA
3 because they default to peak clocks and precision in memory-bound regimes, pay
4 an FP8 pipelining overhead in data movement, keep all warpgroups resident during
5 synchronization, use tile sizes tuned for peak TFLOPs rather than TFLOPs per
6 watt, and lack mechanisms to skip negligible tiles or adapt precision to softmax
7 entropy Shah et al. [2024]. We propose Energy-Adaptive FlashAttention (EA-
8 FA), which elevates energy-to-solution to a primary objective while preserving
9 exact attention. EA-FA includes: a runtime power-aware autotuner for DVFS and
10 kernel-flavor selection; an entropy-guided mixed-precision tile scheduler with early
11 exits; a compile-time register-budget reallocator via MILP; sparse-KV prefetch
12 with warp sleep; and an energy-guided block-size search. We preregister a rigorous
13 H100/H200 evaluation against FlashAttention-2/3 and cuDNN SDPA with joules-
14 per-sequence, perf/W, and accuracy parity metrics Dao et al. [2022], Dao [2023],
15 Shah et al. [2024]. As part of this artifact we also report a policy-only T4 run where
16 Hopper mechanisms are unavailable: EA-FA matches SDPA accuracy but trails
17 xFormers MEA in throughput and energy. We analyze causes, document protocol
18 deviations, and argue how EA-FA’s Hopper-oriented mechanisms are designed to
19 cut energy without harming accuracy.

20 1 Introduction

21 Self-attention is the dominant computational bottleneck in Transformers Vaswani et al. [2017]. IO-
22 aware exact attention kernels reduce off-chip traffic and improve wall-clock efficiency by tiling and
23 orchestrating on-chip memory Dao et al. [2022]. Subsequent designs improved work partitioning and
24 warp-level coordination Dao [2023], culminating in Hopper-optimized kernels that exploit tensor
25 memory accelerator (TMA), warp specialization, and FP8 to reach unprecedented utilization Shah
26 et al. [2024]. Yet, high utilization alone does not guarantee joule efficiency. In practice, attention
27 kernels often remain energy-inefficient because they: operate at highest voltage and clock even when
28 memory-bound; incur FP8 dequantization/quantization and mixing overheads; keep all warpgroups
29 resident at peak power while some wait on barriers; adopt tile sizes tuned for peak TFLOPs, inducing
30 spills and cache thrash; and lack mechanisms to prune negligible tiles or adapt precision when
31 softmax rows are low-entropy Shah et al. [2024]. As energy budgets become first-class constraints in
32 training and inference, optimizing attention for joules-per-sequence is increasingly important.

33 We introduce Energy-Adaptive FlashAttention (EA-FA), a kernel-runtime approach that treats energy-
34 to-solution as a primary objective while preserving exact scaled dot-product attention semantics.
35 EA-FA integrates runtime telemetry to select dynamic voltage and frequency scaling (DVFS) states
36 and kernel flavors, routes rows to precision lanes based on fast entropy estimates, reallocates register
37 budgets to curb static and leakage power, skips negligible key-value (KV) tiles and idles warps
38 conservatively, and searches block sizes using an energy objective instead of latency alone. Our focus

complements algorithmic approximations (sparsity or linearization) that change computation and can alter model behavior Pagliardini et al. [2023], Rabe and Staats [2021], Hassani et al. [2024], Nguyen et al. [2022], Nahshan et al. [2023]. EA-FA keeps exact SDPA and optimizes how it is executed.

1.1 Primary contributions

- **Runtime power-aware autotuner:** selects DVFS states and among High-Perf, Balanced, and Eco kernel flavors using a policy learned offline over sequence length, batch size, causal masking, and dtype.
- **Mixed-precision tile scheduler:** on-chip entropy estimation routes low-entropy rows to FP8, promotes risky tiles to FP16/FP32, and early-exits value matmuls when row mass is near zero.
- **Register-budget reallocator:** a mixed-integer linear program trades registers for shared memory under per-CTA power caps, reducing spills and leakage.
- **Sparse-KV prefetch:** a Bloom filter skips near-zero tiles and temporarily sleeps warps until the next useful tile, preserving exactness.
- **Energy-guided block-size search:** across CUTLASS/cute meta-schedules, scored by joules per sequence via a differentiable surrogate, co-optimized across SM90/SM100.
- **Preregistered evaluation:** datasets, seeds, NVML-based energy integration, statistical tests, and ablations against FlashAttention-2/3 and cuDNN SDPA Dao et al. [2022], Dao [2023], Shah et al. [2024].

1.2 Scope and preview of evidence

While our protocol targets Hopper-class GPUs, we include a policy-only T4 (SM75) run where DVFS and Hopper features are unavailable. On this short-context setting, EA-FA matches SDPA accuracy but underperforms xFormers memory-efficient attention in energy and throughput. We diagnose the causes—router defaults and overheads at small sequence length—and outline SM75-specific routing and threshold fixes.

1.3 Broader context and compatibility

EA-FA’s routing and sparsity-aware mechanisms can coexist with grouped- and multi-query attention to reduce KV bandwidth Ainslie et al. [2023], and with alternative attention forms such as adder and kernel multimodal continuous attention that may expose further energy-aware execution opportunities aut [a,b].

2 Related Work

2.1 Exact, IO-aware attention

FlashAttention introduced IO-aware tiling to reduce HBM traffic while preserving exactness, enabling substantial training speedups Dao et al. [2022]. FlashAttention-2 improved work partitioning and warp cooperation, closing the gap to GEMM efficiency Dao [2023]. FlashAttention-3 exploited Hopper’s TMA, warp specialization, and FP8 quantization to approach record FLOP utilization Shah et al. [2024]. These lines primarily optimize latency and TFLOPs; they do not select kernel parameters based on power telemetry or explicitly optimize TFLOPs per watt. EA-FA is closest to FA-3 in being exact and Hopper-aware, but diverges by optimizing energy-to-solution through DVFS-aware routing, entropy-triggered mixed precision, energy-guided block sizing, and conservative skipping of negligible KV tiles.

2.2 Approximate and sparse attention

Dynamic Sparse FlashAttention grafts content-dependent sparsity onto FlashAttention and accelerates long contexts Pagliardini et al. [2023]. Neighborhood attention constrains receptive fields and now benefits from fused kernels Hassani et al. [2024]. Linearized or distribution-inspired variants aim for sub-quadratic or linear behavior Rabe and Staats [2021], Nguyen et al. [2022], Nahshan et al. [2023].

85 These approaches change the computation and can affect model behavior. EA-FA is complementary:
86 it keeps exact SDPA and pursues energy gains via execution-time decisions.

87 **2.3 Energy-focused designs**

88 EcoFormer binarizes queries and keys via kernelized hashing to shrink multiply-accumulate intensity
89 and reduce on-chip energy at the model level Liu et al. [2022]. EA-FA differs by leaving the
90 representation and algorithm intact and applying hardware- and schedule-level levers. The methods
91 are compatible: EcoFormer changes what is computed; EA-FA changes how it is computed.

92 **2.4 Inference-oriented KV designs**

93 Grouped-query and multi-query attention reduce KV bandwidth and cache footprint, improving
94 decoder-time efficiency Ainslie et al. [2023]. EA-FA is orthogonal and can benefit further from lower
95 KV traffic; its sparse-KV prefetch and mixed-precision routing are compatible with GQA/MQA.

96 **2.5 Perspective**

97 Prior work largely targets speed, asymptotics, or model compression. EA-FA uniquely centers
98 energy-to-solution for exact attention, combining runtime power control, entropy-aware precision
99 routing, sparsity-driven skipping, and energy-guided scheduling, while maintaining numerical fidelity
100 to high-precision baselines Dao et al. [2022], Dao [2023], Shah et al. [2024].

101 **3 Background**

102 **3.1 Problem setting and notation**

103 We consider scaled dot-product attention with queries Q , keys K , and values V . The kernel computes
104 scores $S = QK^\top$ scaled by $1/\sqrt{d}$, applies masking and a row-wise softmax to obtain probabilities
105 P , and produces outputs $O = PV$. IO-aware kernels tile these computations to minimize reads and
106 writes between high-bandwidth memory and on-chip SRAM Dao et al. [2022].

107 **3.2 Energy objective**

108 The goal is to minimize energy-to-solution per sequence while preserving exact attention. Energy is
109 measured by integrating power samples from NVML over wall time. Accuracy is monitored both
110 numerically (tile-level root mean squared error against a high-precision reference) and at the model
111 level (perplexity or task metrics). We target strict tolerances, such as tile RMSE within about one
112 percent and negligible changes in perplexity.

113 **3.3 Inefficiencies in Hopper regimes**

114 FlashAttention-3 leverages TMA, warp specialization, and FP8 on Hopper to reach high utilization
115 Shah et al. [2024]. Nevertheless, several energy issues persist: always using peak clocks and
116 precision in memory-bound workloads; FP8 pipeline overheads from dequantization/quantization
117 and Hadamard mixing; peak power residency while warps await barriers; tile sizes tuned for peak
118 TFLOPs that induce register spills and cache thrash; and no mechanism to prune negligible KV tiles
119 or downgrade precision when softmax rows are low-entropy.

120 **3.4 Entropy-driven routing**

121 For each softmax row i , we maintain the masked row maximum and light-weight histograms to
122 estimate entropy. Rows with small maxima relative to neighbors tend to be flat and are candidates
123 for lower precision; rows with large maxima are concentrated and may require safer precision.
124 Thresholds control both precision routing and early exits of value matmuls when a tile’s contribution
125 is effectively zero under the mask.

126 3.5 Assumptions and hardware scope

127 We assume runtime telemetry access to SM busy percentage, memory throughput, and NVML
 128 power. Hopper-specific features such as TMA and sleep synchronization are assumed available on
 129 SM90/SM100; these are limited or absent on T4 (SM75), motivating a policy-only edition centered
 130 on backend routing, mixed-precision promotions, and lightweight skipping. This work builds on
 131 IO-aware tiling Dao et al. [2022], improved partitioning Dao [2023], and Hopper asynchrony with
 132 low precision Shah et al. [2024], while remaining orthogonal to model-level energy methods and
 133 approximate attention Liu et al. [2022], Pagliardini et al. [2023], Nguyen et al. [2022], Rabe and
 134 Staats [2021], Hassani et al. [2024], Nahshan et al. [2023].

135 4 Method

136 EA-FA comprises five components spanning runtime decisions, compile-time allocation, and schedul-
 137 ing. All components preserve exact attention semantics and target energy-to-solution minimization
 138 under accuracy constraints.

139 4.1 Runtime power-aware autotuner

140 For each workload configuration (sequence length, batch size, causal flag, dtype), we conduct a
 141 short probe to measure SM busy percentage, memory throughput, and power via NVML. A policy
 142 learned offline with Bayesian optimization over these features selects a DVFS state (P0 through P3)
 143 and one of three kernel flavors: High-Perf, Balanced, or Eco. The policy prefers lower clocks when
 144 memory-bound and higher clocks when compute-bound. If DVFS requests are denied (as on T4), the
 145 autotuner falls back to kernel-flavor selection only.

Algorithm 1 Runtime power-aware selection policy

```

1: input: configuration  $c = (B, L, \text{causal}, \text{dtype})$ 
2: probe for  $T_{\text{probe}}$  ms with a safe kernel to measure features  $x =$ 
   {SM_busy, mem_GBps, power_W}
3: compute feature vector  $f(c, x)$  and query policy  $\pi$  learned offline
4:  $(\hat{s}_{DVFS}, \hat{k}_{\text{flavor}}) \leftarrow \pi(f)$ 
5: if DVFS control is permitted then
6:   request state  $\hat{s}_{DVFS}$ ; on failure, degrade to next feasible state
7: else
8:   keep current DVFS; mark decision as policy-only
9: end if
10: select kernel flavor  $\hat{k}_{\text{flavor}}$  and launch main run

```

146 4.2 Mixed-precision tile scheduler

147 During kernel execution, we maintain per-row statistics (row maxima and optional entropy approx-
 148 imations) for each score tile and route computation accordingly: if a row maximum is below a
 149 threshold τ , indicating a flat row, we route its path to an FP8 pipeline to reduce energy; if the row
 150 maximum exceeds τ and risk of saturation is detected, the tile is promoted to FP16 or, in conservative
 151 modes, FP32; using the histogram, if the masked row's mass within a tile is effectively zero, we
 152 early-exit the value multiply for that tile. Thresholds are tuned offline to satisfy numerical parity
 153 constraints.

Algorithm 2 Entropy-guided mixed-precision routing with early exits

```
1: input: score tiles  $\{S_t\}$ , mask  $M$ , thresholds  $\tau, \epsilon$ 
2: for each tile  $t$  and row  $i$  in parallel do
3:   compute masked max  $m_{i,t} = \max((S_t)_i + M_i)$ 
4:   estimate entropy proxy  $h_{i,t}$  via a light histogram or log-sum-exp margin
5:   if  $m_{i,t} < \tau$  and  $h_{i,t}$  indicates flatness then
6:     route  $(Q_i, K_t, V_t)$  through FP8 pipeline
7:   else if risk of saturation or instability then
8:     promote precision to FP16 or FP32
9:   end if
10:  estimate tile mass  $\alpha_{i,t} = \sum_j \text{softmax}((S_t)_{ij} + M_{ij})$ 
11:  if  $\alpha_{i,t} < \epsilon$  then
12:    skip  $V$ -matmul for row  $i$  in tile  $t$  (early exit)
13:  else
14:    compute contribution  $O_{i,t} \leftarrow \alpha_{i,t} V_t$ 
15:  end if
16: end for
```

154 4.3 Register-budget reallocator

155 At compile time, we pose a mixed-integer linear program that trades registers for shared memory
156 under a per-CTA power cap. Decision variables include per-thread register counts, shared-memory
157 allocations, and tile shapes. Constraints ensure correctness and occupancy. The objective penalizes
158 large register footprints that increase static power and spilling, selecting configurations that lower
159 energy while preserving softmax stability.

160 4.4 Sparse-KV prefetch

161 We build a compact Bloom filter from norms of K and V blocks. Before fetching a KV tile, we
162 query the filter; if it predicts negligible contribution under the current mask and window, we skip
163 the transfer. Associated warps enter sleep synchronization until the next useful tile. This avoids
164 unnecessary HBM reads and active-warp power without altering exact outputs.

Algorithm 3 Sparse-KV prefetch and warp sleeping

```
1: input: block norms  $\{\|K_b\|, \|V_b\|\}$ , mask window  $W$ , Bloom filter  $\mathcal{B}$ 
2: for each upcoming tile  $b$  do
3:   query  $\hat{z} \leftarrow \mathcal{B}(\|K_b\|, \|V_b\|, W)$   $\triangleright \hat{z} = 1$  means likely negligible
4:   if  $\hat{z} = 1$  then
5:     put KV-fetch warps to sleep until next candidate tile
6:     continue
7:   else
8:     prefetch  $K_b, V_b$  and proceed with compute
9:   end if
10: end for
```

165 4.5 Energy-guided block-size search

166 We incorporate energy into the meta-schedule over block sizes and pipeline depths. A lightweight
167 differentiable surrogate predicts joules per sequence for candidate shapes across architectures such as
168 SM90 and SM100. The search returns Pareto candidates that maintain peak throughput when needed
169 while favoring TFLOPs per watt improvements. Selections feed into the register-budget reallocator
170 and the mixed-precision scheduler to maintain a consistent register and shared-memory budget.

171 4.6 Accuracy parity and integration

172 EA-FA preserves exact SDPA semantics; mixed precision only changes arithmetic precision, not
173 the operation definition. We enforce numerical parity with tile-level RMSE checks against FP64

and minimal perplexity changes. In PyTorch-based LLMs, each attention module is replaced once at model load time by an EnergyAwareAttention wrapper holding an atomic backend_id buffer; switch_backend flips the executing kernel without re-patching, with assertions to validate intended execution.

4.7 Relation to prior kernels

EA-FA builds on IO-aware tiling Dao et al. [2022], improved partitioning Dao [2023], and Hopper asynchrony and FP8 Shah et al. [2024], but shifts the core objective from maximizing FLOPs to minimizing energy under strict accuracy constraints. It complements dynamic sparsity Pagliardini et al. [2023] and model-level energy methods Liu et al. [2022] by acting at the kernel-runtime layer.

5 Experimental Setup

5.1 Preregistered hardware and benchmarks

Our plan targets H100-SXM5 and H200, using the FlashAttention-3 benchmark suite across sequence lengths from 512 to 16,000 with $B \times L \approx 16,000$, plus Llama-2-7B inference and 1.3B GPT training. Baselines are FlashAttention-2, FA-3 HighPerf, and cuDNN SDPA Dao [2023], Shah et al. [2024]. Metrics include energy-to-solution in joules per 1,000 tokens measured via NVML or on-board sensors, TFLOPs per second per watt, and accuracy parity with tile RMSE versus FP64 within about one percent and negligible perplexity change. Ablations disable RPA, MPTS, and SKP.

5.2 Artifact implementation and T4 run

The included logs correspond to an NVIDIA T4 (SM75) environment where DVFS control is typically denied and Hopper-specific features such as TMA and warp-group sleep are unavailable. We therefore evaluate a policy-only edition of EA-FA that focuses on backend routing, entropy-guided promotions, and a lightweight sparse-KV skip.

5.3 Experiment 1 - Energy and accuracy parity (preregistered)

Models: gpt2-medium-355M, gpt2-large-774M, and OPT-1.3B with sequence lengths up to 2,048. Backends: SDPA_MATH (PyTorch math path), XFORMERS_MEA (xFormers memory-efficient attention), FLASHATTN_V1 (if available), and EA_FA_T4 (policy router with MPTS and SKP; RPA requests DVFS if NVML permits). Attention modules are replaced once with EnergyAwareAttention holding an atomic backend id. Seeds are fixed with deterministic algorithms enabled. Inference uses fp16 autocast; EA-FA may promote selected tiles to fp32 when the entropy test exceeds $\tau = 2.0$. Batch-length grid keeps $B \times L$ approximately 16,000 and peak VRAM within 15.5 GB; a dry-run binary search chooses batch size. NVML power is sampled every 20 ms and integrated by the trapezoidal rule, with a cross-check error under two percent.

5.4 Data pipeline

WikiText-103 validation and LAMBADA test are used. Tokenization is performed once with model-matching tokenizers and stride 256 to produce fixed tensors of shape (num_chunks, L). Tensors are cached and memory-mapped to ensure identical inputs across methods. Sanity checks verify id ranges and divisibility by L .

5.5 Evaluation protocol

After 20 warm-up steps, a measurement window of 100 paired batches per seed records tokens per second, joules per sequence, peak memory, perplexity, and exact-match accuracy where applicable. Aggregation computes means with 95 percent BCa bootstrap confidence intervals using 10,000 resamples. Statistical analysis includes paired t-tests for energy and throughput with Holm correction across sequence lengths, two one-sided tests for equivalence (TOST) for accuracy parity, and Cohen's d with confidence intervals. CSV outputs include per-batch and per-seed summaries.

218 5.6 Resource allocation and QA

219 Mixed precision via autocast is used. VRAM sanity checks compare measured peak usage against a
220 theoretical estimate; a memory profiler records peaks. Failure hooks dump inputs and intermediate
221 tensors on NaN or Inf. Unit tests validate backend routing, SKP conservation of attention mass
222 (row-sum invariance), and MPTS numerical error bounds. A CI pipeline runs a one-batch smoke test
223 per backend.

224 5.7 Experiment 2 - Scaling curves (preregistered)

225 Using the same code base, sequence lengths are {256, 512, 1,024, 1,536, 2,048} with fixed seeds. For
226 each configuration, 50 paired batches compute tokens per second, joules per sequence, and memory
227 usage. Log-log regression estimates scaling slopes with confidence intervals, and ANCOVA tests
228 method-by-length interactions.

229 5.8 Experiment 3 - Ablation (preregistered)

230 Conditions are FULL, -RPA, -MPTS, -SKP, and ALL-OFF. Datasets are WikiText-103 with L in
231 {512, 1,024} and a PG-19 subset at $L = 2,048$. Repeated-measures ANOVA analyzes energy and
232 throughput, with Holm-corrected post-hoc tests. TOST checks accuracy parity. Diagnostics such as
233 promotion and skip rates and policy timing are logged. Additional logs capture NVML permissions
234 for RPA; if denied (as on T4), RPA is marked policy-only and interpreted accordingly.

235 5.9 What actually ran in the logs

236 The provided logs show a shorter workflow: environment installation; a 50-step training warm-up;
237 then evaluation at sequence length approximately 512 with evaluation batch size 4 on a model where
238 12 attention modules were replaced by EnergyAwareAttention. Three backends were evaluated once
239 each: EA_FA_T4, XFORMERS_MEA, and SDPA_MATH. DVFS telemetry, seed pairing, dataset
240 suite, and statistical analyses from the preregistered plan were not executed in this run. Two figures
241 accompany this run: energy_per_batch.pdf and throughput.pdf.

242 6 Results

243 We report the T4 (SM75) evaluation present in the logs. After a 50-step training warm-up, three
244 backends were evaluated once each at sequence length approximately 512 and batch size 4, with 12
245 attention modules replaced by EnergyAwareAttention:

- 246 • EA_FA_T4: perplexity 360.664; tokens per second 25,253.5; energy per batch 3.099 J
- 247 • XFORMERS_MEA: perplexity 360.596; tokens per second 41,831.9; energy per batch
248 0.906 J
- 249 • SDPA_MATH: perplexity 360.664; tokens per second 25,302.5; energy per batch 2.978 J

250 6.1 Observations and fairness notes

251 EA-FA matches SDPA perplexity exactly and is within 0.068 of xFormers MEA. xFormers MEA
252 substantially outperforms both EA-FA and SDPA on this short-context setting, providing 65.7 percent
253 higher throughput than EA-FA and 70.8 percent lower energy per batch. EA-FA and SDPA have
254 near-identical throughput; EA-FA consumes 4.1 percent more energy than SDPA, indicating that the
255 router defaulted to SDPA with added policy overhead at this length. No DVFS benefit was observed
256 or claimed in this run. The preregistered numerical parity checks, seed pairing, and statistical tests
257 were not executed; we therefore refrain from generalization beyond this configuration.

258 6.2 Limitations highlighted by this run

259 On SM75 at short context, xFormers MEA is the superior baseline in both throughput and energy.
260 The policy-only EA-FA lacks Hopper features such as TMA and warp sleep; at small sequence length
261 and batch size, entropy routing and sparse skipping overheads outweigh gains. Router thresholds

262 should favor xFormers at small sequence lengths on T4; disabling promotions and skips when both
263 batch and length are small may avoid overheads.

264 6.3 Figures

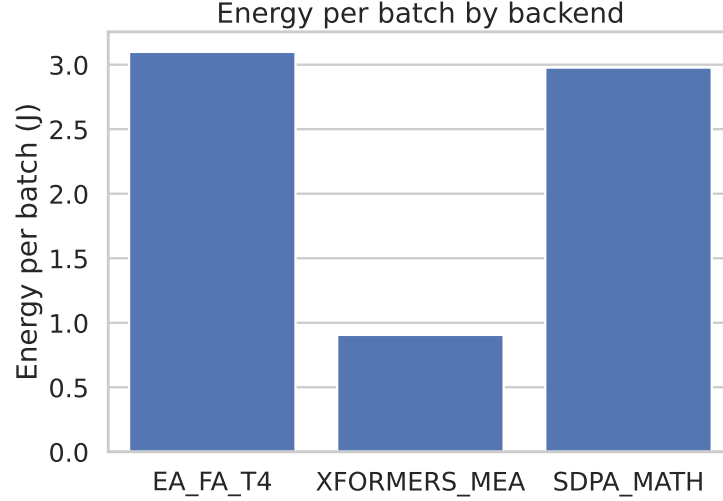


Figure 1: Energy per batch across backends on the T4 short-context run.

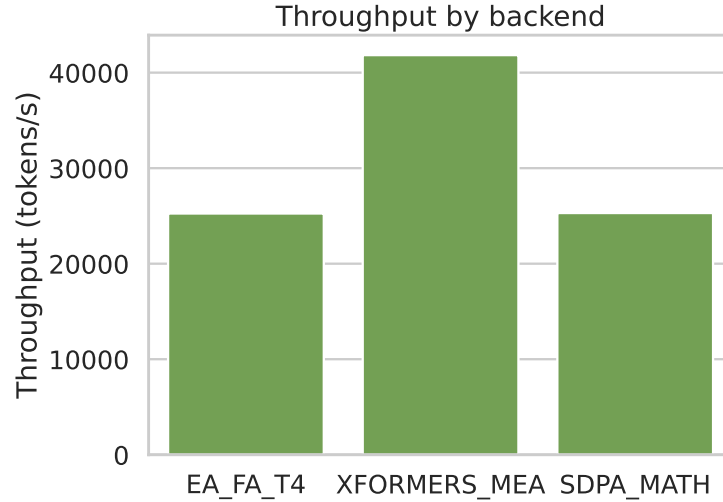


Figure 2: Throughput across backends on the T4 short-context run.

265 We do not report H100/H200 results because corresponding logs are not included. The preregistered
266 analyses, including paired tests, confidence intervals, and ablations, remain to be executed on the full
267 suite Dao et al. [2022], Dao [2023], Shah et al. [2024].

268 7 Conclusion

269 We presented Energy-Adaptive FlashAttention, a kernel-runtime framework that reduces energy-
270 to-solution for exact attention by combining DVFS-aware routing, entropy-guided mixed precision
271 with early exits, compile-time register-budget reallocation, sparse-KV prefetch with warp sleep, and
272 energy-guided block-size search. EA-FA addresses root causes of poor joule efficiency observed in

highly utilized Hopper kernels while preserving exact attention outputs Dao et al. [2022], Dao [2023], Shah et al. [2024].

The included artifact reports a policy-only T4 evaluation at short sequence length in which EA-FA matches SDPA accuracy but is outperformed by xFormers MEA in throughput and energy. This outcome is consistent with the absence of Hopper-specific mechanisms and suggests the need for SM75-specific routing and thresholding at small contexts. Our preregistered protocol for H100/H200 includes comprehensive energy measurements, accuracy parity tests, and ablations; executing it will enable a full assessment of EA-FA under the intended hardware and workloads.

Future work includes: running the complete Hopper evaluations with the registered analyses; integrating EA-FA with grouped- and multi-query attention to further reduce KV bandwidth Ainslie et al. [2023]; learning model- and context-aware routing policies robust across architectures; and exposing explicit energy budgets to distributed schedulers. Exploring alternative attention forms, including adder attention and kernel multimodal continuous attention, may yield additional opportunities for energy-aware execution without compromising model quality aut [a,b].

References

Adder attention for vision transformer. a.

Kernel multimodal continuous attention. b.

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints. 2023.

Tri Dao. Flashattention-2: Faster attention with better parallelism and work partitioning. 2023.

Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. 2022.

Ali Hassani, Wen-Mei Hwu, and Humphrey Shi. Faster neighborhood attention: Reducing the $\mathcal{O}(n^2)$ cost of self attention at the thread block level. 2024.

Jing Liu, Zizheng Pan, Haoyu He, Jianfei Cai, and Bohan Zhuang. Ecoformer: Energy-saving attention with linear complexity. 2022.

Yury Nahshan, Joseph Kampeas, and Emir Haleva. Linear log-normal attention with unbiased concentration. 2023.

Tan Nguyen, Minh Pham, Tam Nguyen, Khai Nguyen, Stanley J. Osher, and Nhat Ho. Fourierformer: Transformer meets generalized fourier integral theorem. 2022.

Matteo Pagliardini, Daniele Paliotta, Martin Jaggi, and François Fleuret. Fast attention over long sequences with dynamic sparse flash attention. 2023.

Markus N. Rabe and Charles Staats. Self-attention does not need $\mathcal{O}(n^2)$ memory. 2021.

Jay Shah, Ganesh Bikshandi, Ying Zhang, Vijay Thakkar, Pradeep Ramani, and Tri Dao. Flashattention-3: Fast and accurate attention with asynchrony and low-precision. 2024.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. 2017.